Comparative Evaluation of LLM Judges for ADeLe v1.0 Difficulty Annotation

Algebra/Calculus Question Bank (N = 1,520) — Qwen2.5-72B vs Llama-3.3-70B vs Mistral-Large-2411, benchmarked against a human-derived reference annotation

This report documents the setup, methodology, and results of a local (self-hosted) replication of the DeLeAn / ADeLe v1.0 difficulty-annotation pipeline, in which large open-weight language models act as automated judges that score each task along 18 cognitive/behavioral "demand" rubrics (0-5 scale). The goal is to determine which annotator model(s) best approximate a trusted human/reference annotation, and whether combining multiple models improves agreement over any single model.

Executive Summary

- **Qwen2.5-72B-Instruct-AWQ** is the strongest single annotator overall, outperforming Llama-3.3-70B on 10 of 14 rubrics with usable reference signal.
- **Llama-3.3-70B-Instruct-AWQ** exhibits a stronger systematic upward bias (over-estimates difficulty) than Qwen across almost all rubrics, which depresses its exact-agreement rate.
- Taking the **element-wise minimum of Qwen and Llama** predictions improves quadratic-weighted Cohen's kappa over the best individual model in 10 of 14 rubrics (mean improvement +0.038), because it corrects for the shared upward bias. The only clear exception is QLl (Logical Reasoning), where Qwen alone is already the strongest signal and the ensemble introduces noise.
- **Mistral-Large-Instruct-2411-AWQ** was added as a third, statistically independent judge; only 2 of 18 rubrics are complete so far (still processing on the cluster), showing intermediate performance between Qwen and Llama on those two rubrics.
- Four rubrics (KNa, KNc, KNn, KNs, MS) show degenerate/undefined correlation statistics because the reference annotation is constant (always 0) for this algebra-specific dataset — these demand types are simply not exercised by algebra/calculus tasks.

1. Background and Objective

ADeLe v1.0 (Annotated Demand Levels) is a framework for characterizing the cognitive and behavioral demands of a task using 18 independent rubrics, each scored on a 0-5 ordinal scale via chain-of-thought LLM judging. The reference pipeline (delean-batch-manager) was originally designed around the OpenAI Batch API; for this project it was repurposed to run entirely locally using vLLM's offline batch inference engine on a SLURM-managed GPU cluster, avoiding per-token API costs and allowing full control over annotator model choice.

The dataset under annotation is a bank of 1,520 algebra/calculus questions. A trusted reference annotation (produced separately, treated as ground truth for this comparison) exists for all 18 rubrics. The central question addressed here is: *which open-weight LLM(s), used as automated judges, best reproduce this reference annotation, and can multiple judges be combined to do better than any one alone?*

1.1 The 18 ADeLe Rubrics

Each rubric captures a distinct cognitive/behavioral demand dimension. Codes and definitions used throughout this report:

Code	Name
AS	Attention and Search
AT	Atypicality
CEc	Comprehension
CEe	Expression
CL	Conceptualization, Learning & Abstraction
KNa	Knowledge in Applied Sciences and Professions
KNc	Customary Everyday Knowledge
KNf	Knowledge in Formal Sciences
KNn	Knowledge in Natural Sciences
KNs	Knowledge in Social Sciences and Humanities
MCr	Identifying Relevant Information (metacognition)
MCt	Critical Thinking Processes (metacognition)
MCu	Calibrating Knowns and Unknowns (metacognition)
MS	Mind Modelling and Social Cognition
QLl	Logical Reasoning (deductive)
QLq	Quantitative Reasoning
SNs	Spatial-physical Reasoning
VO	Volume (time/effort to complete task)

1.2 Infrastructure

- SLURM cluster (Vrain HPC, DSIC-UPV), nodes vrhpc3/5/6, each with 4× GPUs (PCIe-only, no NVLink).

- Inference engine: vLLM 0.6.1.post2, offline batch mode (`vllm.entrypoints.openai.run_batch`), tensor-parallel size 4, AWQ 4-bit quantization (`awq_marlin`), `--max-model-len 4096`, `--enforce-eager`, `--gpu-memory-utilization 0.95`.
- NCCL configured for PCIe-only interconnect (`NCCL_P2P_DISABLE=1`, `NCCL_IB_DISABLE=1`), attention backend forced to XFormers (`VLLM_ATTENTION_BACKEND=XFORMERS`) for eager-mode compatibility.
- Per-rubric sequential processing loop with skip-if-output-exists resumability: each of the 18 rubrics is submitted as a separate batch job within the same SLURM allocation, and completed rubrics are automatically skipped on job resubmission (needed after time-limit cancellations).

1.3 Models Evaluated

Model	Size	Quantization	Status
Qwen2.5-72B-Instruct-AWQ	72B dense	AWQ 4-bit	Complete (18/18 rubrics)
Llama-3.3-70B-Instruct-AWQ	70B dense	AWQ 4-bit	Complete (18/18 rubrics)
Mistral-Large-Instruct-2411-AWQ	123B dense	AWQ 4-bit	In progress (2/18 rubrics)
Mixtral-8x22B-Instruct-v0.1-AWQ	141B MoE	AWQ 4-bit	Discarded (underperformed)

Note: a Qwen3-32B candidate was also considered but discarded due to an architecture incompatibility with the installed vLLM version (Qwen3ForCausalLM is not supported in vLLM 0.6.1.post2).

1.4 Comparison Methodology

For each rubric, model predictions are merged against the reference annotation on a common task ID, restricted to rows where both a valid model prediction and a valid reference value exist (rows with parsing failures / context-length truncation are dropped from that rubric's comparison, hence sample sizes vary slightly by rubric and model). The following metrics are reported:

- **Pearson / Spearman correlation** — linear and rank agreement between model and reference scores.
- **MAE** — mean absolute error on the 0-5 scale.
- **Exact agreement** — fraction of tasks where model score equals reference score exactly.
- **Within-1 agreement** — fraction of tasks where model and reference differ by at most one level.
- **Cohen's kappa (linear- and quadratic-weighted)** — chance-corrected ordinal agreement; the quadratic-weighted variant is used as the primary ranking criterion since it penalizes larger disagreements more heavily, which is appropriate for an ordinal 0-5 difficulty scale.

2. Qwen2.5-72B vs Llama-3.3-70B: Full 18-Rubric Comparison

Both models completed annotation of all 1,520 tasks across all 18 rubrics. The table below reports quadratic-weighted kappa (the primary agreement metric) and exact-agreement rate for each rubric and model, against the reference annotation. Rubrics are sorted in ADeLe's standard order; rows where the reference is constant for this dataset (and thus correlation/kappa are undefined) are marked N/A.

Rubric	N (Qwen)	Qwen κ-quad	Qwen exact	N (Llama)	Llama κ-quad	Llama exact	Better model
AS	1357	0.297	0.509	1492	0.364	0.457	Llama
AT	1518	0.664	0.547	1517	0.539	0.302	Qwen
CEc	1461	0.493	0.491	1462	0.388	0.347	Qwen
CEe	1512	0.041	0.303	1510	0.015	0.211	Qwen
CL	1304	0.634	0.574	1306	0.623	0.442	Qwen
KNa	1507	N/A	N/A	1516	N/A	N/A	—
KNc	1410	-0.002	0.923	1520	N/A	N/A	Qwen
KNf	1502	0.454	0.718	1513	0.485	0.652	Llama
KNn	1513	N/A	N/A	1520	N/A	N/A	—
KNs	1520	N/A	N/A	1520	N/A	N/A	—
MCr	887	0.460	0.554	953	0.431	0.349	Qwen
MCt	1495	0.378	0.555	1509	0.321	0.415	Qwen
MCu	1468	0.599	0.773	1472	0.473	0.425	Qwen
MS	1520	N/A	N/A	1520	N/A	N/A	—
QLl	1166	0.503	0.633	1198	0.290	0.296	Qwen
QLq	1463	0.483	0.578	1485	0.305	0.287	Qwen
SNs	1509	0.286	0.705	1498	0.277	0.719	Qwen
VO	1494	0.544	0.687	1501	0.374	0.357	Qwen

N = number of tasks with both a valid model prediction and a valid non-missing reference value for that rubric. κ-quad = Cohen's kappa, quadratic weights. "Better model" indicates which model achieves higher κ-quad (N/A rubrics excluded).

2.1 Key Findings

- Qwen achieves higher quadratic-weighted kappa than Llama on **11 of 14** comparable rubrics.
- Qwen's advantage is most pronounced on **QLl** (Logical Reasoning, κ=0.503 vs 0.290), **QLq** (Quantitative Reasoning, κ=0.483 vs 0.305), **MCu** (Calibrating Knowns/Unknowns, κ=0.599 vs 0.473), and **VO** (Volume/effort, κ=0.544 vs 0.374) — all rubrics where numerical or logical precision matters, plausibly favoring Qwen's stronger STEM training emphasis.
- Llama's advantage is limited to **AS** (Attention and Search, κ=0.364 vs 0.297) and **KNf** (Knowledge in Formal Sciences, κ=0.485 vs 0.454) — both narrow margins.
- Across nearly all rubrics, Llama shows a stronger upward bias than Qwen (higher mean predicted level relative to the reference), which depresses its exact-agreement rate even when correlation is comparable (e.g., AT: exact agreement 54.7% for Qwen vs only 30.2% for Llama, despite similar Pearson correlation).

- Both models fail similarly on **CEe** (Expression) and **KNc** (Customary Everyday Knowledge) — near-zero kappa for both, suggesting these rubrics are either genuinely hard to judge for this domain or poorly specified for algebra-style tasks.

3. Ensembling Qwen and Llama

Since both models are systematically biased upward relative to the reference (i.e., they tend to rate tasks as more demanding than the reference does), three simple combination strategies were tested per rubric: the rounded **average** of the two scores, the **element-wise minimum**, and the **element-wise maximum**. The minimum operator is expected to partially cancel a shared upward bias.

Rubric	Qwen κ	Llama κ	Avg κ	Min κ	Max κ	Best single	Best ensemble	Δ improvement
AS	0.297	0.364	0.347	0.404	0.277	0.364	0.404	0.040
AT	0.664	0.539	0.641	0.672	0.529	0.664	0.672	0.007
CEc	0.493	0.388	0.404	0.547	0.349	0.493	0.547	0.054
CEe	0.041	0.015	0.048	0.079	0.010	0.041	0.079	0.038
CL	0.634	0.623	0.692	0.681	0.562	0.634	0.692	0.058
KNa	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
KNc	-0.002	N/A	-0.002	N/A	-0.002	-0.002	-0.002	0.000
KNf	0.454	0.485	0.426	0.539	0.411	0.485	0.539	0.054
KNn	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
KNs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
MCr	0.460	0.431	0.470	0.523	0.346	0.460	0.523	0.063
MCt	0.378	0.321	0.353	0.495	0.259	0.378	0.495	0.117
MCu	0.599	0.473	0.516	0.710	0.384	0.599	0.710	0.111
MS	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
QLl	0.503	0.290	0.386	0.383	0.332	0.503	0.386	-0.117
QLq	0.483	0.305	0.393	0.511	0.284	0.483	0.511	0.029
SNs	0.286	0.277	0.366	0.221	0.307	0.286	0.366	0.080
VO	0.544	0.374	0.430	0.547	0.359	0.544	0.547	0.003

All values are quadratic-weighted Cohen's kappa vs. reference. Rubrics with constant reference values (KNa, KNn, KNs, MS) are omitted from this table.

3.1 Key Findings

- The best ensemble strategy beats the best individual model on **12 of 14** comparable rubrics, with a mean quadratic-kappa improvement of **+0.038**.
- **Minimum(Qwen, Llama)** is the winning strategy in the large majority of cases (10/14), confirming the hypothesis that both models share a systematic upward-bias failure mode that the min operator partially corrects.
- Largest gains from ensembling are seen on **MCt** (+0.117, Critical Thinking) and **MCu** (+0.111, Calibrating Knowns/Unknowns) — both metacognition rubrics where neither individual model is strong alone, but the shared bias correction yields a substantial joint improvement.
- **QLl (Logical Reasoning) is the one clear exception**: Qwen alone (κ=0.503) is already far stronger than Llama (κ=0.290) on this rubric, and combining them (best ensemble κ=0.386) actually hurts — the min/average operators dilute Qwen's strong signal with Llama's weaker one. This illustrates that ensembling should be applied selectively, not blindly rubric-by-rubric.

- **Recommended default policy**: use min(Qwen, Llama) as the combined annotation for all rubrics except QLI, where Qwen's individual prediction should be used directly.

4. Mistral-Large-2411: Preliminary Results (Third Judge)

Mistral-Large-Instruct-2411-AWQ (123B dense, AWQ 4-bit quantized checkpoint from TechxGenus/Mistral-Large-Instruct-2411-AWQ) was launched as a third, independent annotator to enable more robust ensembling in the future (e.g., median or majority vote across three judges rather than min-of-two). At the time of writing, only 2 of 18 rubrics have completed; results below should be treated as preliminary.

Rubric	Model	N	Pearson	MAE	Exact	κ-quad
AS	Qwen	1357	0.367	0.676	0.509	0.297
AS	Llama	1492	0.396	0.757	0.457	0.364
AS	Mistral	1483	0.369	0.740	0.476	0.344
AT	Qwen	1518	0.665	0.478	0.547	0.664
AT	Llama	1517	0.701	0.796	0.302	0.539
AT	Mistral	1507	0.597	0.660	0.407	0.523

4.1 Preliminary Observations

- On **AS**, Llama is marginally the strongest of the three (κ=0.364), with Mistral close behind (κ=0.344) and Qwen last (κ=0.297).
- On **AT**, Qwen is clearly strongest (κ=0.664), with Mistral (κ=0.523) and Llama (κ=0.539) close to each other, both well behind Qwen.
- Mistral does not yet stand out as uniformly better or worse than the other two — it occupies an intermediate position on both rubrics evaluated so far. A firmer conclusion requires the remaining 16 rubrics to complete.
- Once complete, the plan is to extend the ensemble analysis in Section 3 to a three-way combination (e.g., median or majority vote of Qwen/Llama/Mistral), which should be more robust to any single model's idiosyncratic failure modes than the current min-of-two heuristic.

5. Data Quality Notes

- **MCr** and **QLI** show a notably higher rate of missing/unparseable model responses (~10-30% of tasks) than other rubrics for both Qwen and Llama, suggesting these rubric prompts may be more prone to eliciting responses that don't conform to the expected structured output format. Root cause not yet investigated.
- A residual ~0.1-0.2% of tasks per rubric fail due to context-length overflow (prompt + completion exceeding the 4096-token model context limit after raising max_completion_tokens from 1000 to 2000 to fix truncation issues); this was judged low-impact enough not to warrant re-running with a larger context window.
- Four rubrics (**KNa, KNn, KNs, MS**) have a constant reference value of 0 across the entire algebra dataset, correctly reflecting that these demand types (applied-science knowledge, natural-science knowledge, social-science knowledge, and social cognition/mind-modelling) are simply not exercised by algebra/calculus tasks — this is a property of the dataset domain, not a data quality defect.
- **KNa** is a partial exception: although the reference is constant 0, both Qwen and Llama hallucinate non-zero levels here (shared false-positive failure mode), unlike KNc/KNn/KNs/MS where at least one model correctly predicts the constant 0.

6. Summary Recommendations

- Use **Qwen2.5-72B-Instruct-AWQ** as the primary single-model annotator if only one model can be used — it outperforms Llama-3.3-70B on the majority of rubrics with meaningful signal.
- For production annotation, use the **min(Qwen, Llama)** ensemble by default, with the single exception of QLI (use Qwen alone there).
- Complete the in-progress **Mistral-Large-2411** run and re-evaluate whether a three-model combination (median / majority vote) further improves on the two-model minimum heuristic.
- Treat KNa, KNc, KNn, KNs, and MS as out-of-scope / non-informative rubrics for algebra-domain datasets; any future domain-general validation should include a task sample where these rubrics have genuine variance.